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COURSE CODE: SBT-DF202
LAB NAME: USB IMAGE ACQUISITION AND HASH VERIFICATION
DATE: 29/08/2026
INSTRUCTOR: AMINU IDRIS

Laboratory instructions

Before beginning this lab, review the **Module 3 lessons and study materials** covering forensic disk imaging, USB acquisition, FTK Imager, dd, dcfldd, and hash verification.

Use only a **personal or instructor-approved test USB drive**. Do not use a third-party, organisational, or production device. Carefully identify the USB before acquisition because selecting the wrong device with a low-level imaging tool can result in data loss.

Part A - Prepare the USB Evidence

Create a folder on the test USB named:

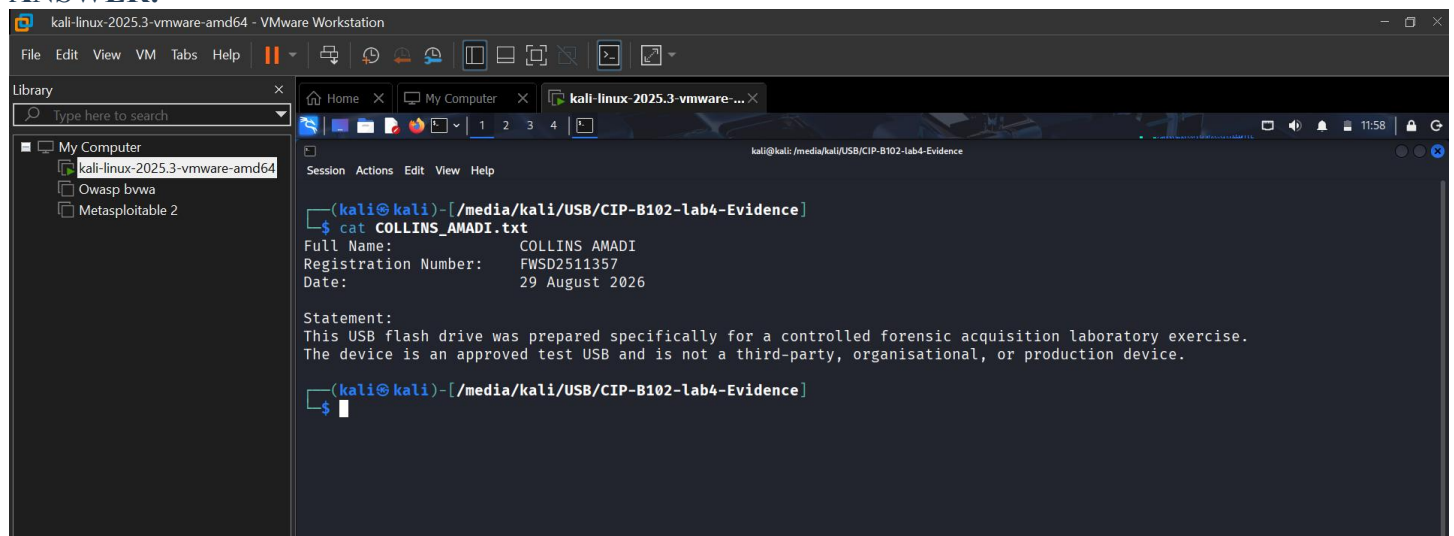
CIP-B102-Lab4-Evidence

Inside the folder, create:

firstname_lastname.txt

The file must contain your full name, registration number, date, and a statement confirming that the USB was prepared for a controlled forensic acquisition lab.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2025.3-vmware-amd64
Owasp bwwa
Metasploitable 2
Home My Computer kali-linux-2025.3-vmware-...
kali@kali: /media/kali/USB/CIP-B102-lab4-Evidence
Session Actions Edit View Help
(kali@kali)-[/media/kali/USB/CIP-B102-lab4-Evidence]
$ cat COLLINS_AMADI.txt
Full Name: COLLINS AMADI
Registration Number: FWSD2511357
Date: 29 August 2026
Statement:
This USB flash drive was prepared specifically for a controlled forensic acquisition laboratory exercise.
The device is an approved test USB and is not a third-party, organisational, or production device.
(kali@kali)-[/media/kali/USB/CIP-B102-lab4-Evidence]
$
```

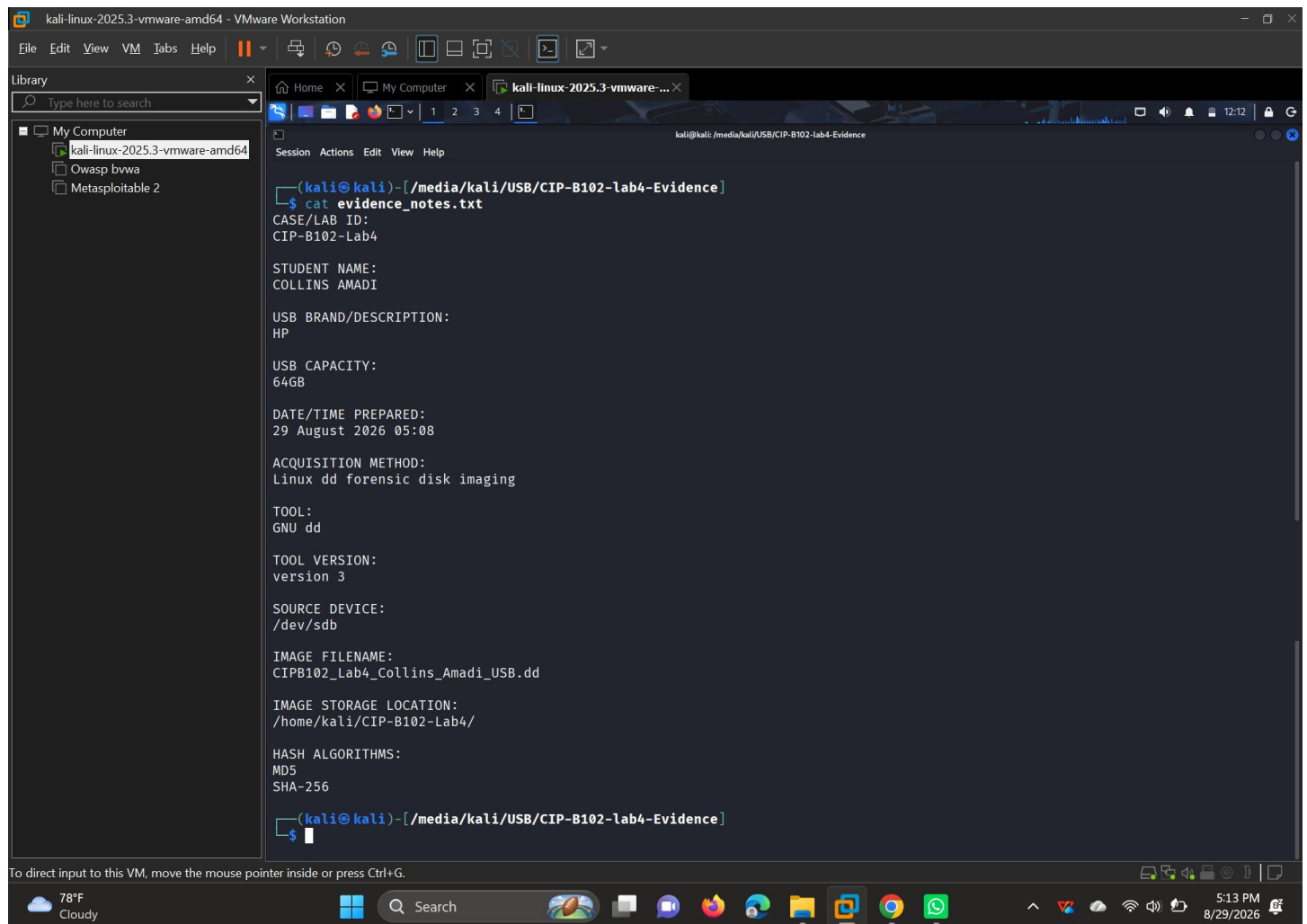
You may also add one image and one PDF file to provide additional file types for validation.

Part B - Complete Evidence Documentation

Before acquisition, record:

- Case/Lab ID
- Student name
- USB brand/description
- USB capacity
- Date and time prepared
- Acquisition method
- Tool and version
- Image filename
- Image storage location
- Hash algorithm used

ANSWER:



```
kali@kali: /media/kali/USB/CIP-B102-lab4-Evidence
$ cat evidence_notes.txt
CASE/LAB ID:
CIP-B102-Lab4

STUDENT NAME:
COLLINS AMADI

USB BRAND/DESCRIPTION:
HP

USB CAPACITY:
64GB

DATE/TIME PREPARED:
29 August 2026 05:08

ACQUISITION METHOD:
Linux dd forensic disk imaging

TOOL:
GNU dd

TOOL VERSION:
version 3

SOURCE DEVICE:
/dev/sdb

IMAGE FILENAME:
CIPB102_Lab4_Collins_Amadi_USB.dd

IMAGE STORAGE LOCATION:
/home/kali/CIP-B102-Lab4/

HASH ALGORITHMS:
MD5
SHA-256

kali@kali: /media/kali/USB/CIP-B102-lab4-Evidence
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

78°F Cloudy 5:13 PM 8/29/2026

Part C - Acquire the USB Image

Complete **at least one** approved acquisition method.

Option A - Windows using FTK Imager

Use FTK Imager to create a forensic image of the USB. Carefully verify the selected device, use Raw/DD where applicable, record evidence information, save the image to your computer rather than the source USB, and enable verification after imaging.

Option B - Linux/Kali using DD

Identify the USB using lsblk or fdisk, unmount the USB partition if necessary, and acquire the whole USB device using an appropriately constructed dd command.

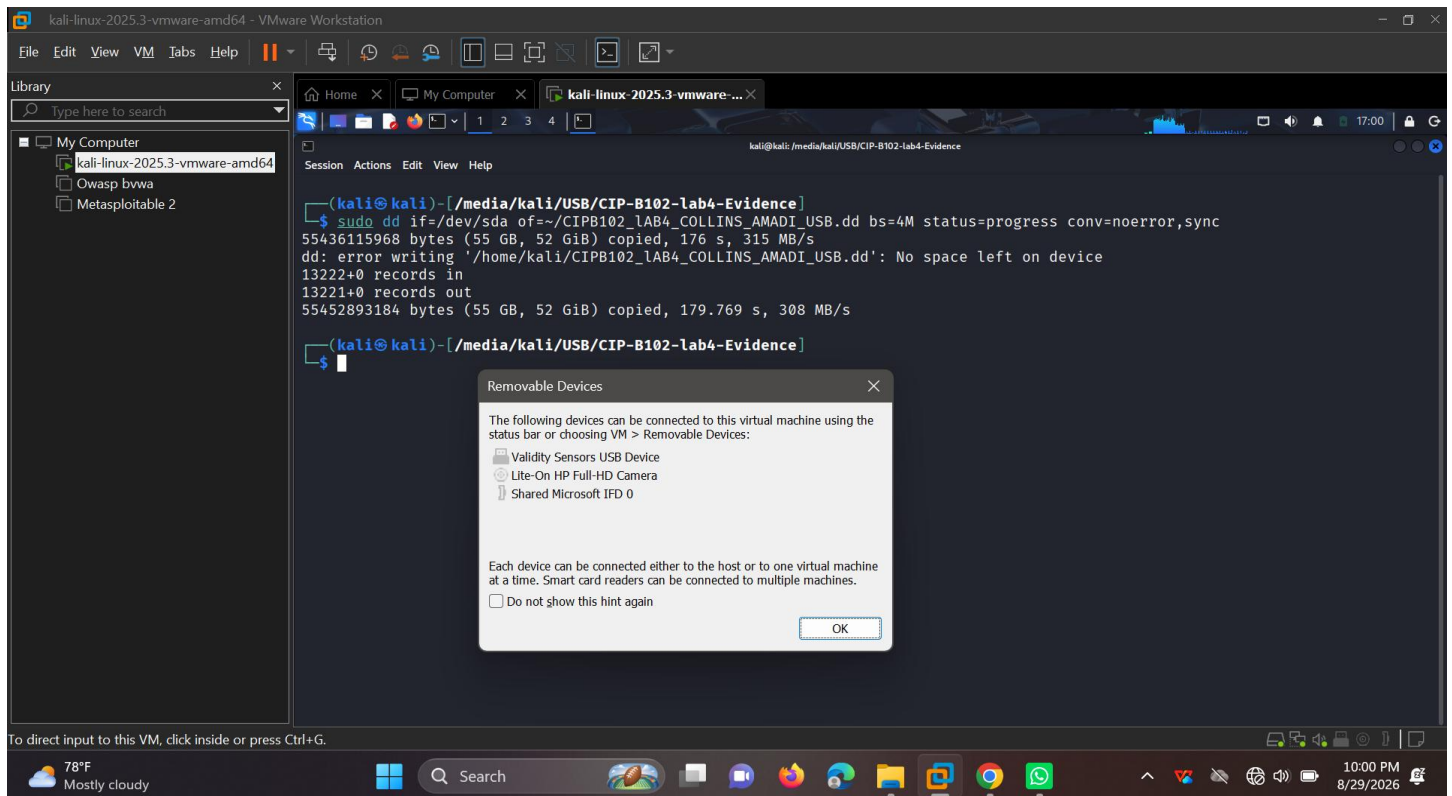
Example structure:

```
sudo dd if=/dev/sdX of=~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd bs=4M status=progress conv=noerror,sync
```

Replace /dev/sdX with the actual USB device identified on your system. **Do not copy the example device name blindly.**

dcfldd may also be used where available.

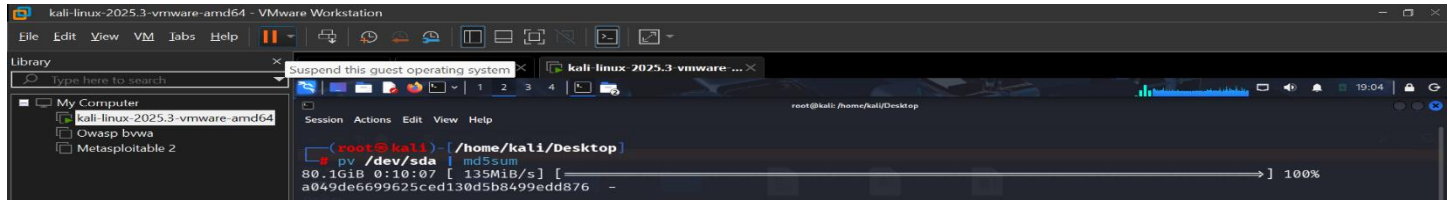
ANSWER:



Part D - Hash Verification

Calculate MD5 and SHA-256 hashes for both the original source USB and the acquired image.

ANSWER:



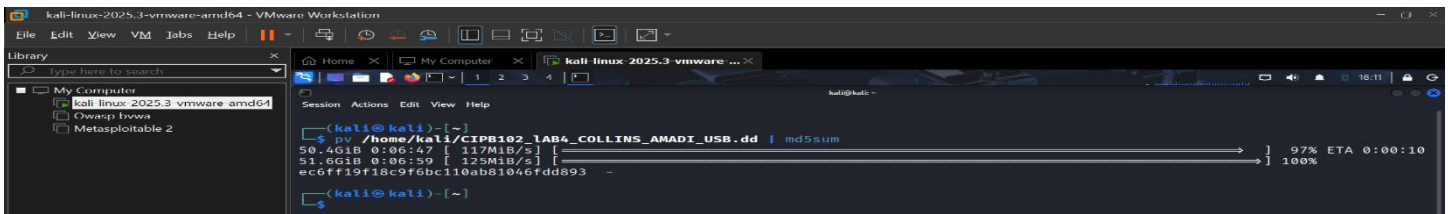
```
kali@kali: ~/home/kali/Desktop
root@kali:~/home/kali/Desktop
# pv /dev/sda | md5sum
80.1GiB 0:10:07 [ 135MiB/s] [ ] 100%
a049de6699625ced130d5b8499edd876 -

# sha256sum
80.1GiB 0:10:52 [ 72.4MiB/s] [ ] 100%
27be2b48ee47559ae7dcfb3388401f6759ee56ece27fe9b09270ccbe1e623a01 -
```



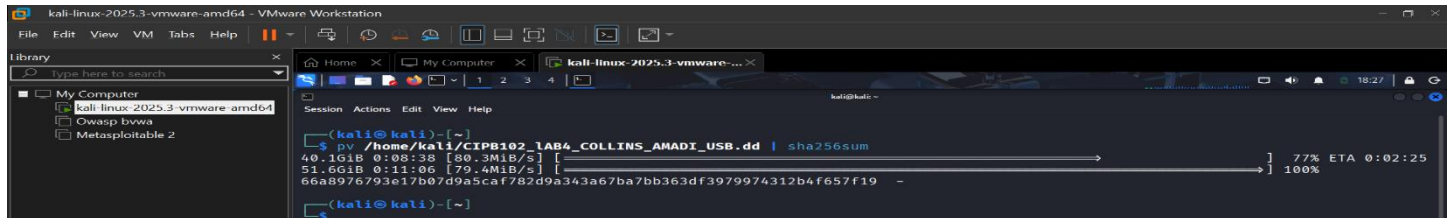
```
kali@kali: ~/home/kali/Desktop
# md5sum
50.4GiB 0:06:47 [ 117MiB/s] [ ] 97% ETA 0:00:10
51.6GiB 0:06:59 [ 125MiB/s] [ ] 100%
ec6ff19f18c9f6bc110ab81046fdd893 -

# sha256sum
40.1GiB 0:08:38 [ 80.3MiB/s] [ ] 77% ETA 0:02:25
51.6GiB 0:11:06 [ 79.4MiB/s] [ ] 100%
66a8976793e1707d9a5caf782d9a343a67ba7bb363df3979974312b4f657f19 -
```



```
kali@kali: ~/home/kali/Desktop
# md5sum
50.4GiB 0:06:47 [ 117MiB/s] [ ] 97% ETA 0:00:10
51.6GiB 0:06:59 [ 125MiB/s] [ ] 100%
ec6ff19f18c9f6bc110ab81046fdd893 -

# sha256sum
40.1GiB 0:08:38 [ 80.3MiB/s] [ ] 77% ETA 0:02:25
51.6GiB 0:11:06 [ 79.4MiB/s] [ ] 100%
66a8976793e1707d9a5caf782d9a343a67ba7bb363df3979974312b4f657f19 -
```



```
kali@kali: ~/home/kali/Desktop
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ec6ff19f18c9f6bc110ab81046fdd893 -

# sha256sum
40.1GiB 0:08:38 [ 80.3MiB/s] [ ] 77% ETA 0:02:25
51.6GiB 0:11:06 [ 79.4MiB/s] [ ] 100%
66a8976793e1707d9a5caf782d9a343a67ba7bb363df3979974312b4f657f19 -
```

Compare the values and document whether they match. If they do not match, explain the possible cause and repeat the acquisition if required.

ANSWER:

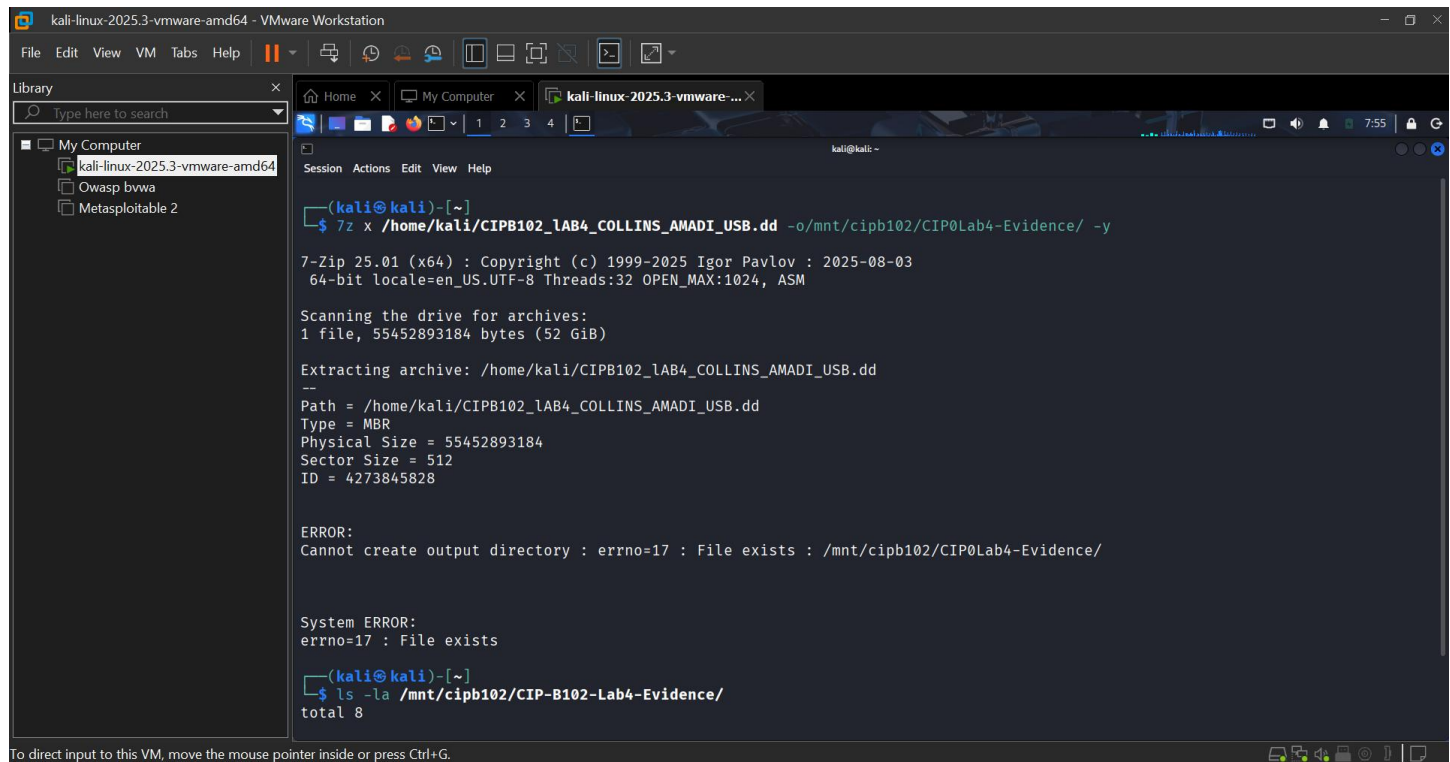
Part E - Validate the Acquired Image

Confirm that your acquired image contains `firstname_lastname.txt`.

You may validate the image using either:

- **FTK Imager**, by adding the .dd image as an evidence item and locating the file, or
- **Linux read-only mounting**, where supported.

ANSWER:



```
kali@kali: ~  
$ 7z x /home/kali/CIPB102_LAB4_COLLINS_AMADI_USB.dd -o/mnt/cipb102/CIP0Lab4-Evidence/ -y  
7-Zip 25.01 (x64) : Copyright (c) 1999-2025 Igor Pavlov : 2025-08-03  
64-bit locale=en_US.UTF-8 Threads:32 OPEN_MAX:1024, ASM  
  
Scanning the drive for archives:  
1 file, 55452893184 bytes (52 GiB)  
  
Extracting archive: /home/kali/CIPB102_LAB4_COLLINS_AMADI_USB.dd  
--  
Path = /home/kali/CIPB102_LAB4_COLLINS_AMADI_USB.dd  
Type = MBR  
Physical Size = 55452893184  
Sector Size = 512  
ID = 4273845828  
  
ERROR:  
Cannot create output directory : errno=17 : File exists : /mnt/cipb102/CIP0Lab4-Evidence/  
  
System ERROR:  
errno=17 : File exists  
  
kali@kali: ~  
$ ls -la /mnt/cipb102/CIP-B102-Lab4-Evidence/  
total 8
```

The source lab requires students to demonstrate that the expected evidence file is actually present in the forensic image.